following from each theory is absolutely identical. (There are two types of holes in a given Calabi-Yau shape, but it turns out that string vibrational patterns—and hence physical implications—are sensitive only to the *difference* between the number of holes of each type. So if one Calabi-Yau has, say, two holes of the first kind and five of the second, while another Calabi-Yau has five holes of the first kind and two of the second, then even though they differ as geometrical shapes, they can give rise to identical physics.[44])

From another perspective, then, this bolsters the suspicion that space is not a foundational concept. Someone describing the universe using one of the five string theories would claim that space, including the extra dimensions, has a particular size and shape, while someone else using one of the other string theories would claim that space, including the extra dimensions, has a different size and shape. Because the two observers would simply be using alternative *mathematical* descriptions of the same *physical* universe, it is not that one would be right and the other wrong. They would both be right, even though their conclusions about space— its size and shape—would differ. Note too, that it's not that they would be slicing up spacetime in different, equally valid ways, as in special relativity. These two observers would fail to agree on the overall structure of spacetime itself. And that's the point. If spacetime were really fundamental, most physicists expect that everyone, regardless of perspective—regardless of the language or theory used—would agree on its geometrical properties. But the fact that, at least within string theory, this need not be the case, suggests that spacetime may be a secondary phenomenon.

We are thus led to ask: if the clues described in the last two sections are pointing us in the right direction, and familiar spacetime is but a large-scale manifestation of some more fundamental entity, what is that entity and what are its essential properties? As of today, no one knows. But in the search for answers, researchers have found yet further clues, and the most important have come from thinking about black holes.

Wherefore the Entropy of Black Holes?